

JAISAL MEHTA

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EDUCATION

Indiana University – Bloomington

May 2025

Bachelor of Science in Computer Science

Minors: Business, Data Science, Systems Engineering

Courses: Object Oriented Programming, Discrete Structures, Data Structures and Algorithms, Systems Programming in Unix, iOS Application Development, Computer Structures, Advanced Algorithms, Introduction to Computer Networks, Introduction to Data Analysis and Mining, Data Representation, Introduction to Statistical Inference, The Computer in Business

PROFESSIONAL EXPERIENCE

Serve IT, Bloomington, IN

August 2023 – December 2023

UX Design Intern

- Collaborated in **Agile** sprints to redesign the Tandem Birthing Center website using **WordPress**, applying **wireframes**, **mockups**, and **prototypes** built in **Figma**, affecting the entire Bloomington, Indiana community
- Conducted **usability testing** and user interviews to improve **information flow** and **user engagement** across devices

Viant Medical, Wheeling, IL

June 2021 – July 2021

Manufacturing Engineering Intern

- Created **validation reports** for medical manufacturing machines, ensuring oil changes met product quality standards towards customer demands and production requirements, while maintaining a clean **Oracle database**
- Utilized **Minitab** to generate **statistical reports**, ensuring product quality through measurement and compliance

CERTIFICATIONS

- NVIDIA AI for All: From Basics to GenAI Practice**, Nvidia Academy October 2025
- AWS Educate Introduction to Generative AI**, Amazon Web Services September 2025
- AWS Educate Machine Learning Foundations**, Amazon Web Services September 2025
- AWS Educate Introduction to Cloud 101**, Amazon Web Services August 2025

TECHNICAL PROJECTS

TasteSearch | Swift, SwiftUI, SwiftData, Google Places API, Gemini AI

January 2025

- Developed a **full-stack** iOS app using **SwiftUI** and **SwiftData**, integrating **Google Places API** and **Gemini AI** to deliver intelligent review analysis, personalized dish recommendations, and **natural language search processing**
- Built a robust backend with **secure API handling**, **Keychain storage**, and API call rate limiting, along with a custom UI system using glass-style components, smooth animations, and haptic feedback

SubTracker | MongoDB, Express.js, Angular, Node.js, Chart.js

December 2025

- Developed a **full-stack** subscription management platform with **JWT-based user authentication** to help users securely track recurring subscriptions, and calculate monthly/yearly spending through centralized management
- Built responsive dashboard with **Chart.js visualizations** displaying spending trends and cost distribution, featuring a **modular Angular component library** (modals, toast notifications, skeleton loaders) with dark/light theme support

Stock Price Prediction with AI/ML | Python, Scikit-learn, TensorFlow, Jupyter Notebooks

June 2025

- Developed and compared **predictive models (Linear Regression vs. LSTM)** on real time stock data using **Python**, **scikit-learn**, and **TensorFlow**; analyzed accurate short-term forecasts and evaluated performance with **MAE and RMSE metrics**
- Applied **deep learning** sequence modeling with **LSTM** networks to capture temporal dependencies in stock price data

Brain Tumor Classification | Python, OpenCV, Scikit-learn

March 2025

- Built and evaluated multiple machine learning models (**K-NN**, **Decision Trees**, **Random Forest**, **CNN**) on **3,000+** brain MRI images, achieving **up to 85% accuracy** in detecting tumor types including glioma, meningioma, and pituitary
- Preprocessed and labeled 3,000+ MRI scans using OpenCV and PIL (resize/normalize + dimensionality reduction for models), enabling consistent inputs and faster experimentation across traditional ML and deep learning pipelines

Simple Data Pipeline | Python, MySQL, REST APIs

October 2024

- Developed a **Python-based ETL pipeline** to store real-time weather data from a **WeatherAPI** to a **SQL database**, along with **shell** and **crontab scripts** in order to **automate ETL pipeline execution** in 15-minute intervals
- Modularized **data parsing**, **cleaning**, **transformation**, and **SQL execution** for an **unlimited** number of data entries

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, C, SQL, Swift, R, HTML, CSS, JavaScript, Spring Boot, React, Angular, Node.js

Tools & Platforms: GitHub, MongoDB, MySQL Workbench, Jupyter, Postman, IntelliJ, PyCharm, Xcode, Unix CLI

Machine Learning & Data Science: Supervised/Unsupervised learning, Regression, Decision Tree, K-NN, SVM, Time Series

Concepts & Practices: Data Structures and Algorithms, RESTful APIs, Agile Development, Version Control (Git), Cloud concepts